SULIT

FACULTY OF COMPUTING
UTM Johor Bahru

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 2 2023/2024

SUBJECT CODE	**:**	**SECJ 2253**
SUBJECT NAME	**:**	**REQUIREMENTS ENGINEERING & SOFTWARE MODELING**
SECTION	**:**	**01, 02, 03, 04, 05**
TIME	**:**	
DATE/DAY	**:**	
VENUES	**:**	

INSTRUCTIONS:

SECTION A :	**10 MULTIPLE CHOICE QUESTIONS**	**(20 MARKS)**
SECTION B :	**9 STRUCTURED QUESTIONS**	**(80 MARKS)**
TOTAL		**(100 MARKS)**

(Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **TWELVE (12)** printed pages excluding this page.

SECTION A: MULTIPLE CHOICE QUESTION [20 MARKS]

Choose the **BEST** answer for the question. Write the answer in the given booklet.

1. What an actor should be or do in a use case diagram?
   A. Could be inside the system boundary.
   B. Use the system inactively.
   C. Must have a role taken by an external system.
   D. Interact directly with the system.

2. What items should NOT be in a use case specification?
   A. Actor
   B. Pre-condition
   C. Flow
   D. Post-condition

3. Figure 1 shows a process shopping order UML activity diagram. Which statement is FALSE related to Figure 1?
   A. Send Invoice and Ship Order take place concurrently at the fork.
   B. Ship Order and Accept Payment should end at the same time before it can be proceeded to the next activity.
   C. Invoice is an activity from Send Invoice.
   D. Fill Order occurs when the order from Receive Order is accepted.

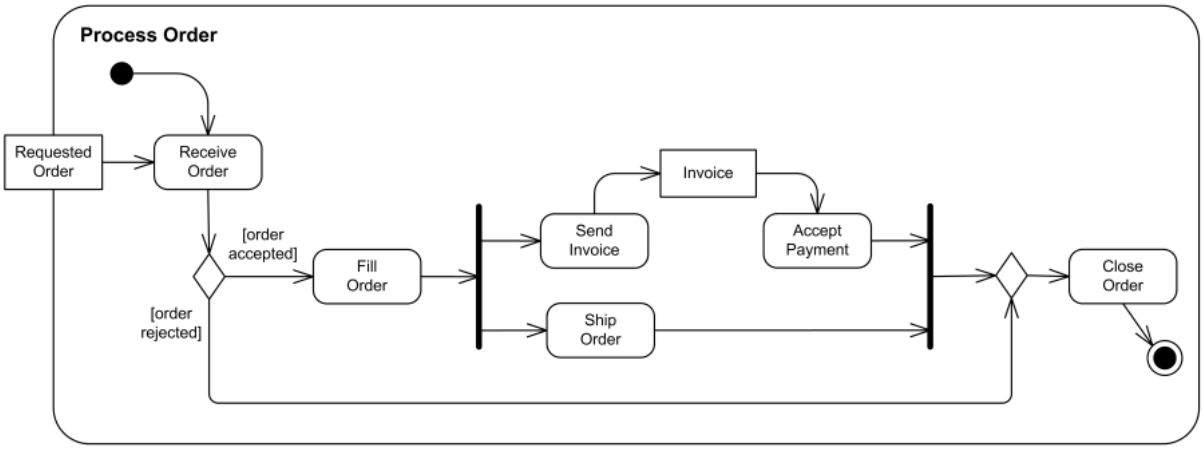


Figure 1 : Shopping order activity diagram

4. Figure 2 shows a domain model diagram. What is the type of relationship as labelled in X?

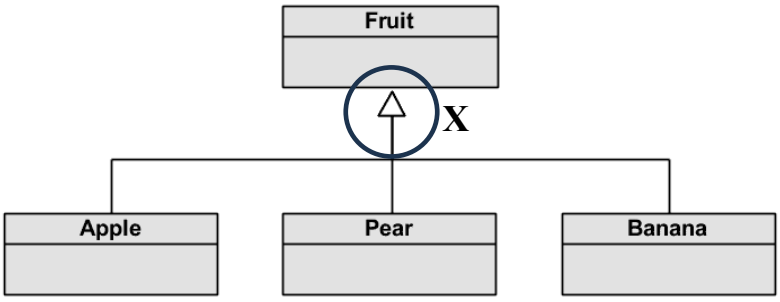


Figure 2: Fruit domain model diagram

A. Association
B. Inheritance
C. Composition
D. Aggregation

5. Which of the following is a shared quality criteria for single requirements AND requirements specification?
   A. Feasible
   B. Complete
   C. Implementation free
   D. Traceable

6. Which of the following is NOT an advantage of documenting system requirements using natural language?
   A. Good readability of requirements
   B. Universal applicability to document any circumstance
   C. Prior knowledge requirement
   D. Absence of formal notation

7. Which of the following system activities is NOT considered relevant when constructing requirements statements using the templates/patterns?
   A. User interaction
   B. Autonomous system activity
   C. Interface requirement
   D. Development constraints

8. Which validation technique thoroughly reviews requirements by a group of experts according to specific criteria?
   A. Inspection
   B. Prototyping
   C. Walkthrough
   D. Commenting

9. Which option BEST describes the validation of requirements in terms of their content quality aspect, concerning whether they accurately represent the wishes and needs of stakeholders?
   A. Traceability
   B. Completeness
   C. Correctness
   D. Consistency

10. Which aspect of traceability helps in the validation and verification of system requirements?
    A. Tracing requirements to design elements and test cases
    B. Tracing requirements to project schedules and budgets
    C. Tracing requirements to stakeholder feedback
    D. Tracing requirements to project documentation

SECTION B: STRUCTURED QUESTION **[80 MARKS]**

There are NINE (9) questions. The marks for each part of the question are as indicated. Write the answer in the given booklet.

Question 1 **[6 marks]**

Figure 3 shows the use case diagram for a reporting system that involves two actors only. List THREE (3) errors and explain how to correct the errors.

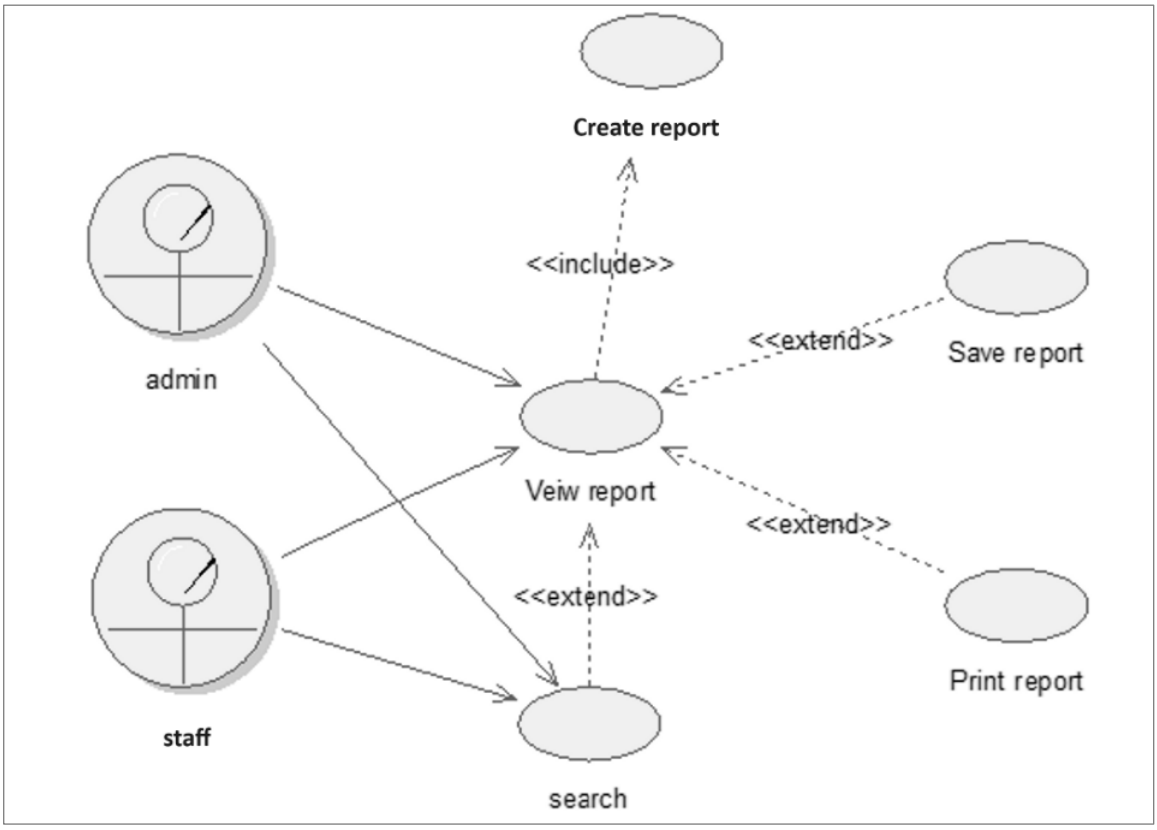


Figure 3: Use case diagram for a reporting system

Question 2 **[5 marks]**

Nice Café plans to give discounts to their loyal customers who dine in or order for takeaways at their café. Thus, the manager requests TechByte Sdn. Bhd. to upgrade the current Point-of-Sale (POS) system so that it enables the café staff to register memberships of their customers using handphone numbers. Upon paying their ordered food via POS system, the staff will request customers who have not registered as a member to share their names and phone number. If customers agree to be a member, the staff will enter the customers' details via the system. The system checks whether the customers exist before it can register using the unique phone numbers.

Based on the provided scenario regarding the POS system development for Nice Café, analyze the requirements and finalize the use case specification by fulfilling items (i) through (v) outlined in Table 1.

Table 1: Use case specification for UC001 Register membership

Use Case ID	**UC001**
Use Case Name	Register membership
Description	The staff creates a new registration for a customer.
Actor	**(i)**
Pre-condition	The customer must… **(ii)**
Normal Flow	1. The staff… **(iii)** 2. The system… **(iv)** 3. The system creates a registration with the customers' details if the registration does not exist (see 3a) 4. The system prompts the successful registration
Alternative Flow	3a. If the customer already exists, the system prompts the message that the a new registration cannot be created.
Exception Flow	1. The system displays an error message if the network connection is lost 2. Return to normal flow
Post-condition	**(v)**

Question 3 **[6 marks]**

Figure 4 presents a domain class model for university teaching. Identify and describe FOUR (4) relationships between the classes, including their multiplicities, as depicted in the model.

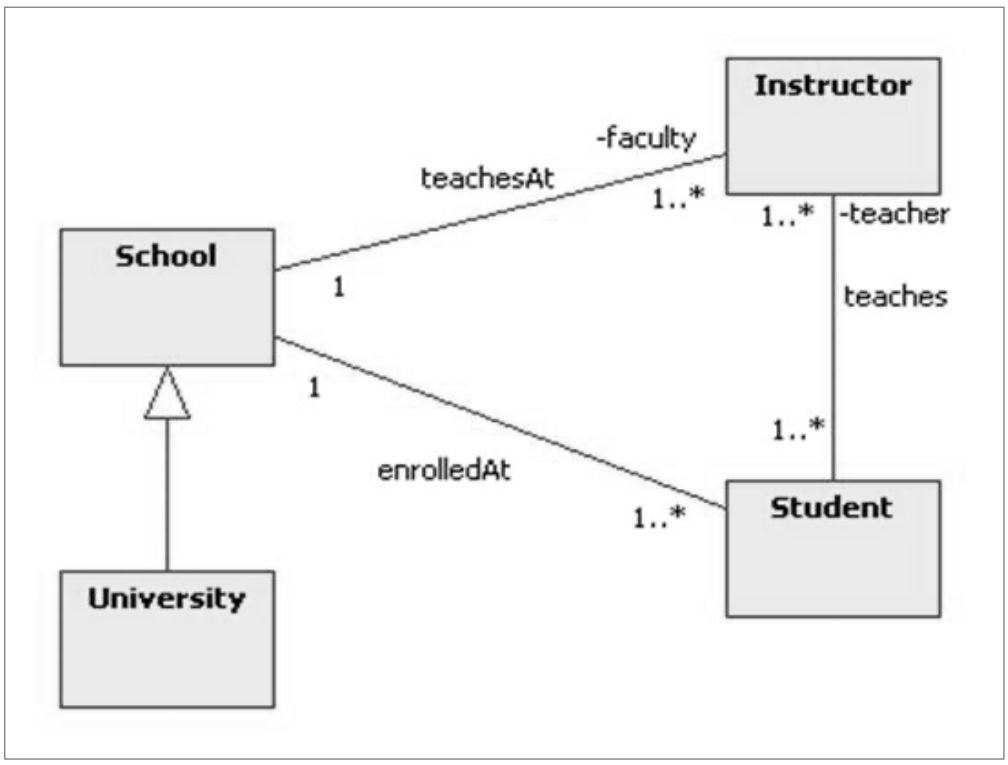


Figure 4: University teaching domain model diagram

Question 4 **[5 marks]**

Based on the following descriptions for conducting a course within a faculty, analyze the scenario and complete the state transition diagram by providing the missing items (i) to (iv) as shown in Figure 5.

A course will exist when the term started and terminated or no longer proceeded when it is being cancelled. As the term started, the course is being taught at the faculty, and as the classes end, the course will be having its Final Exams, which then will be terminated when it is closed. While the course is being taught, if the number of student is dropped to zero, the course will no longer be offered / no longer be in existence. Meanwhile, if the number of student is dropped, but there is/are still a few students left, the course will still be ongoing and be being taught.

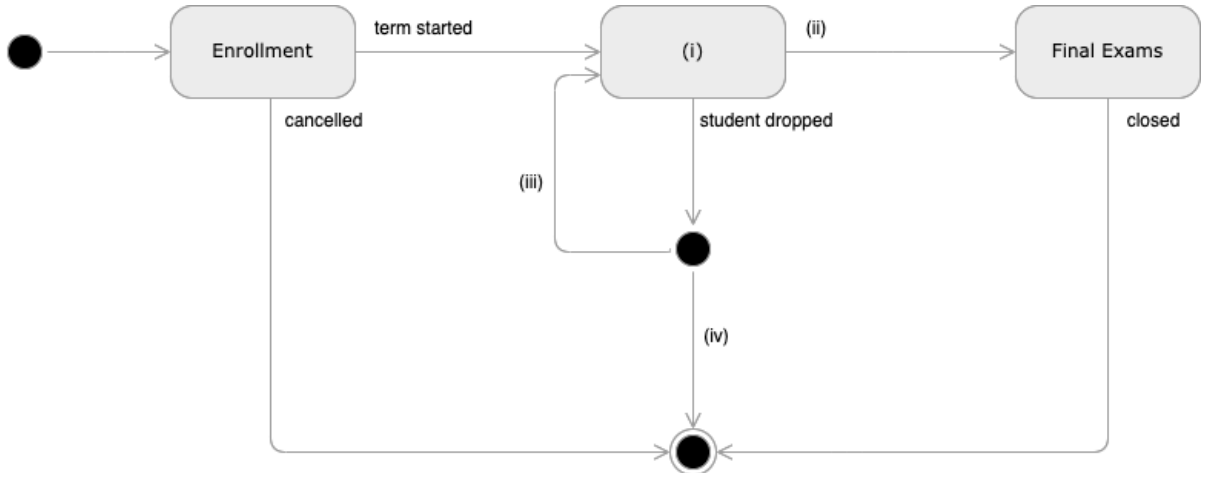


Figure 5: State transition diagram for running a course in faculty

Question 5 **[4 marks]**

Table 2 lists the requirements statements that do not meet certain quality criteria. Identify the quality criteria that each requirement does not meet.

Table 2: Requirements versus quality criteria

ID	Requirement Statements	Quality Criteria
REQ1	The system shall only accept password with specific symbols.	
REQ2	The system shall include the tax for all products.	
REQ3	The system shall use a dialog box to display a warning message.	
REQ4	The system shall accept huge number of entries during the peak hours.	

Question 6 **[10 marks]**

Based on the description, verify the three requirements in Table 3 against the desired quality of requirements (clear, focus, correct, feasible, etc.). Report the findings using the format provided in Table 4.

The Green Haven Nursery (GHN) company, guided by the project consultant from Unicorn Software House (USH), aims to enhance nursery management through a management information system (MIS). GHN company currently getting the nursery products such as plants, flowers, soils, fertilizers, gardening tools, equipment, and machinery from various products suppliers.

The proposed MIS must handle storing and updating customer information, tracking purchases, and managing loyalty programs. The system must integrate with third-party platforms or services that manage customer data analysis, personalized marketing, and loyalty program management. Also, the proposed system must consider the integration with financial institutions that offer online payment gateway to streamline the improvised purchasing process.

It should also offer extensive inventory management features, enabling staff to monitor stock levels, track plant varieties, and receive alerts for low stock items. The system must ensure accurate recording and tracking of inventory, sales transactions, and customer information. It is important to prevent errors in inventory management, sales records, and customer interactions, thereby maintaining data integrity and trustworthiness. The system should be user-friendly and intuitive, allowing staff members to perform tasks efficiently with minimal training. Furthermore, the system must adhere to strict budgetary constraints. Additionally, it should be developed within a specified timeframe to meet operational deadlines. Furthermore, the system must be compatible with existing hardware and software infrastructure to minimize disruptions during implementation.

Table 3: Requirements for MIS

Requirement ID	Requirements
RQ1	The system shall be able to allow staff members to perform tasks efficiently with minimal training.
RQ2	The system shall be able to track purchases and generate report at the end of the month.
RQ3	The system must be integrated with online payment gateway Alipay to streamline the improvised purchasing process.

Table 4: Example of report findings for MIS requirements problems

Requirement ID	Problem Type	Problem Description	Recommendation
Req. ID	*the problem/mistake by the author*	*description of the problem/mistake*	*Your suggestion to solve the problem/mistake*

Question 7 **[14 marks]**

Based on the above Green Haven Nursery (GHN) company scenario in Question 6, answer the following questions (a) until (c):

(a) Discuss TWO (2) techniques that the USH consultant may consider to avoid ambiguity in documenting the requirements using natural language. (4 marks)

(b) Analyze the types of transformation effects in the following requirements statements in Table 5. Justify your reason. (6 marks)

Table 5: Transformation Effects in MIS Requirements

Requirement ID	Requirements
REQ-a	Customer should be able to register all membership information required.
REQ-b	The data shall be displayed to the staff member on the point-of-sale terminal.
REQ-a	To track the order purchases, the users need to input the data

(c) Improve requirements statements in Table 6 using the appropriate requirement template/pattern as shown in Figure 6. (4 marks)

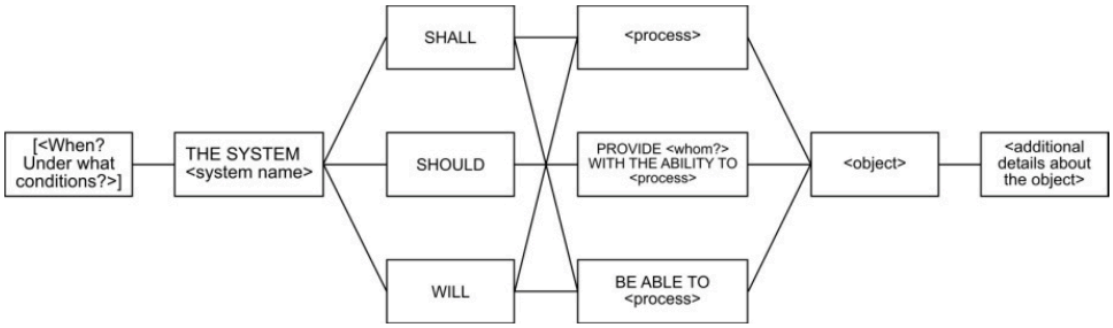


Figure 6: Requirements Template

Table 6: Transformation Effects in MIS Requirements

Requirement ID	Requirements
REQ-d	Customer should be able to register all membership information required.
REQ-e	The data shall be displayed to the staff member on the point-of-sale terminal.

Question 8 **[14 marks]**

The Green Haven Nursery (GHN) company aims to enhance nursery management through a management information system (MIS). Figure 7 illustrates the analysis of the proposed nursery-MIS presented by the project consultant from Unicorn Software House (USH) following the requirements elicitation activities.

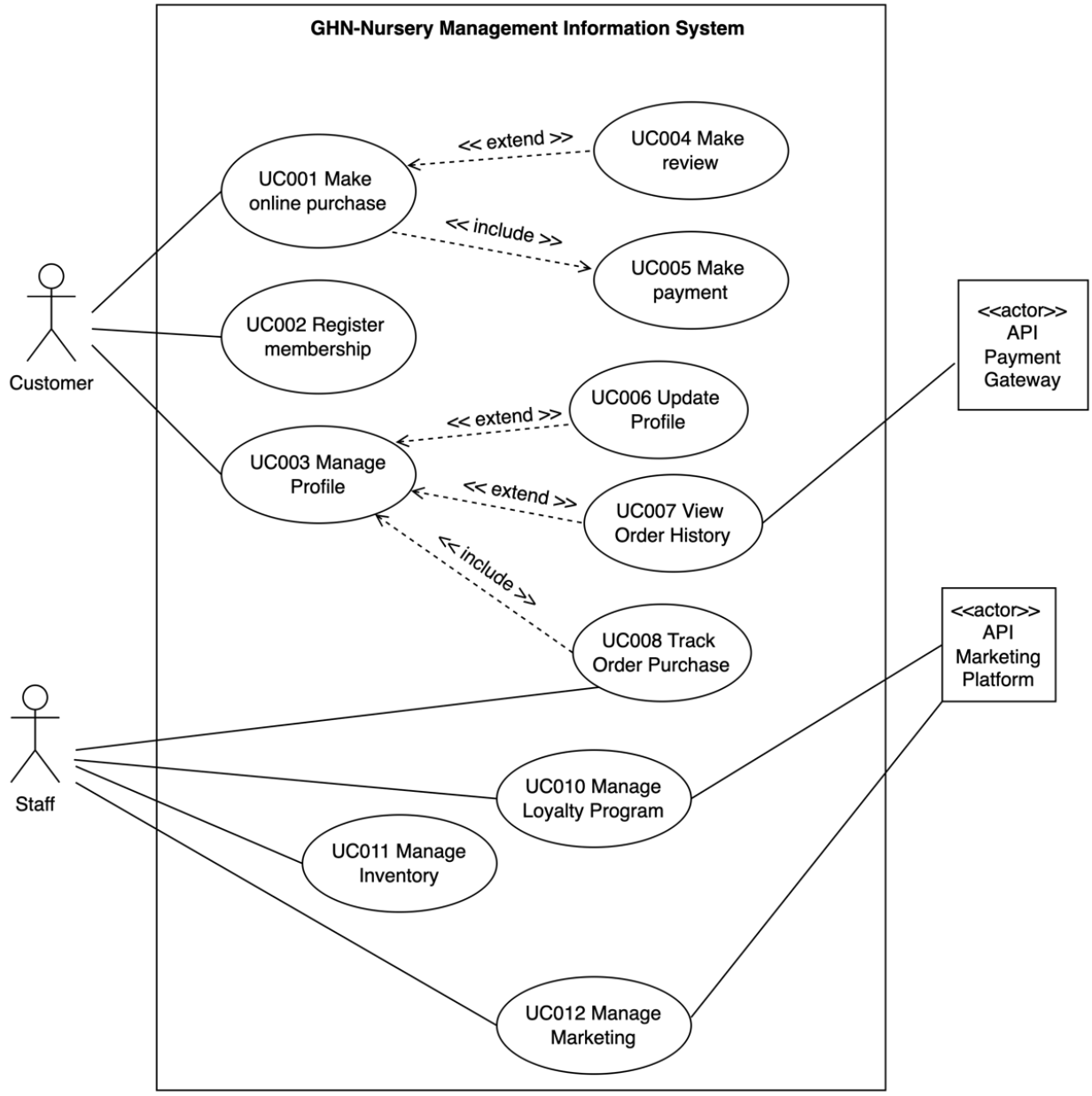


Figure 7: UCD #1 Use case diagram of proposed nursery MIS

Following that, Table 7 presents a portion of the review report provided by the inspection team during the requirements validation process.

Table 7: Review findings by inspection team

No	No/Page	Reviewer Comment	Reviewer	Type of findings
1.	UCD #1	Use case 'manage customer account' and <<actor>> System Admin is missing	Representative of client team (GHN company)	**Minor defect**
2.	UCD #1	Relationship between use case 'make payment' and <<actor>> API payment gateway system is missing	Representative of consultant (USH developer team)	**Undefined**
3.	UCD #1	The tracking of order purchases exhibits unclear flows. In the correct scenario, while managing their profile, customers can choose to update their profile, view their order history, or opt to track their current order purchase.	Representative of client team (GHN company)	**Open issue**

Based on the above Figure 7 and Table 7, answer the following questions (a) until (e):

(a) Analyze the type of findings and suggest replacing "**Minor defect**" in review finding number **1** in Table 7. Explain your reason for the answer. (4 marks)

(b) Analyze the type of findings and suggest replacing "**Undefined**" in review finding number **2** in Table 7. Explain your reason for the answer. (3 marks)

(c) Analyze the type of findings and suggest replacing "**Open Issue**" in review finding number **3** in Table 7. Explain your reason for the answer. (3 marks)

As the proposed nursery MIS for GHN company is developed, conflicts arise among stakeholders due to differing perspectives, priorities, and interpretations of requirements. Identify the conflicts based on the given scenario in (d) and (e). Justify your answer. (4 marks)

(d) The project manager proposes open-source technologies for cost reduction, but the head of IT security opposes it, preferring closed-source technologies for better security. This conflict stems from differing favours regarding technology choices and risk management.

(e) The marketing manager insists on implementing real-time personalized marketing in the MIS towards improving customer engagement and sales. While the IT manager argues that it may compromise system performance and highlighting a priority towards maintaining system efficiency and reliability.

Question 9 **[16 marks]**

Active Tutoring Center (ATC), a prominent provider of personalized tutoring services, employs a versioning system to manage updates and revisions during requirements engineering. Using version control software, they assign unique version numbers to each requirement, enabling precise tracking of changes, from small adjustments to significant updates. This ensures seamless collaboration and adaptability in meeting evolving educational needs.

(a) Describe THREE (3) categories for classifying change requests by the ATC stakeholders during requirements management. (6 marks)

(b) Table 8 outlines various scenarios involving change requests from different ATC stakeholders, affecting the corresponding requirements in the current version. For each scenario, propose an updated version number and justify your suggestion. (10 marks)

Table 8: Stakeholder change requests and requirements version

Scenario	Version
i. A student using the study materials document finds a section confusing and requests clarification on a particular topic. The curriculum development team reviews the request and agrees to add additional explanations to address the student's confusion.	`07.02.10`
ii. A parent request the system to integrate new educational technologies, such as interactive simulations or virtual reality experiences, into the tutoring platform.	`10.02`
iii. A stakeholder suggests integrating a new subject, such as coding or robotics, into the tutoring curriculum to align with emerging educational trends.	`02.01.05`
iv. A parent notices that a hyperlink in the study materials document is broken and submits a change request to update it with the correct link.	`04.03`
v. A change in administrative staff prompts a request to update contact information in the tutoring materials.	`05.02.05`

- END OF QUESTIONS -